

Activity

Group members:

Working with count data

The data in this class activity comes from a study on residents of Framingham, MA, which was conducted to research variables related to heart health. We will work with a subset of the data, containing

- **cigsPerDay**: The number of cigarettes smoked per day during the study period.
- **education**: 1 = High School, 2 = Some College, 3 = College Degree, 4 = Advanced Degree.
- **male**: 1 = Male, 0 = Female.
- **age**: The age of the individual in years.
- **diabetes**: 1 if the individual has diabetes, 0 otherwise.
- **BMI**: the individual's body mass index (BMI)
- **currentSmoker**: 1 if the individual currently smokes, 0 otherwise

Below is output of a quasi-Poisson model using this data. Use this output to answer the following questions.

```
smokers <- heart_data |>
  filter(currentSmoker == 1)

m2 <- glm(cigsPerDay ~ male + age + education + diabetes + BMI,
          data = smokers, family = quasipoisson)
summary(m2)
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.837192	0.115099	24.650	< 2e-16	***
male	0.456170	0.026095	17.481	< 2e-16	***
age	-0.006806	0.001595	-4.267	2.07e-05	***
education2	0.016463	0.029675	0.555	0.579	
education3	0.016427	0.037706	0.436	0.663	
education4	-0.015033	0.040477	-0.371	0.710	
diabetes	-0.025398	0.092606	-0.274	0.784	
BMI	0.005001	0.003304	1.513	0.130	

(Dispersion parameter for quasipoisson family taken to be 5.519388)

```
vcov(m2)
```

	(Intercept)	male	age	education2
(Intercept)	1.324781e-02	-7.543261e-05	-1.209099e-04	-1.108129e-03
male	-7.543261e-05	6.809567e-04	-1.457262e-06	6.767626e-05
age	-1.209099e-04	-1.457262e-06	2.543831e-06	1.004946e-05
education2	-1.108129e-03	6.767626e-05	1.004946e-05	8.806037e-04
education3	-9.653811e-04	1.224285e-04	6.075923e-06	4.199471e-04
education4	-9.018688e-04	-2.755796e-05	6.764320e-06	4.043217e-04
diabetes	7.665450e-04	4.907976e-05	-9.513520e-06	-1.651884e-05
BMI	-2.656023e-04	-1.295274e-05	-1.397067e-07	7.396988e-06

	education3	education4	diabetes	BMI
(Intercept)	-9.653811e-04	-9.018688e-04	7.665450e-04	-2.656023e-04
male	1.224285e-04	-2.755796e-05	4.907976e-05	-1.295274e-05
age	6.075923e-06	6.764320e-06	-9.513520e-06	-1.397067e-07
education2	4.199471e-04	4.043217e-04	-1.651884e-05	7.396988e-06
education3	1.421765e-03	3.929524e-04	-1.153424e-05	7.973004e-06
education4	3.929524e-04	1.638377e-03	-1.992754e-05	8.305380e-06
diabetes	-1.153424e-05	-1.992754e-05	8.575954e-03	-1.940554e-05
BMI	7.973004e-06	8.305380e-06	-1.940554e-05	1.091919e-05

Questions

1. Is there a difference in the average number of cigarettes smoked between smokers with some college education and smokers with a college degree, after accounting for sex, age, diabetes, and BMI?